Data-Driven Traffic Insights for Local Council Road Planning (London)

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Introduction

This project evaluates road-traffic patterns in London by four datasets AADF, Count Points, Local Authority Traffic, and MRDB (2018-2024) provided by the Department for Transport (DfT). The study uses a combination of geospatial, temporal, and machine learning techniques to detect long-term trends, hotspots, and post pandemic recovery trends in the capital over the whole area. The methodology follows the best practice of transport analytics, in which the multi-year, spatially-referenced datasets are necessary to provide strong mobility detection (Tzika-Kostopoulou et al., 2024).

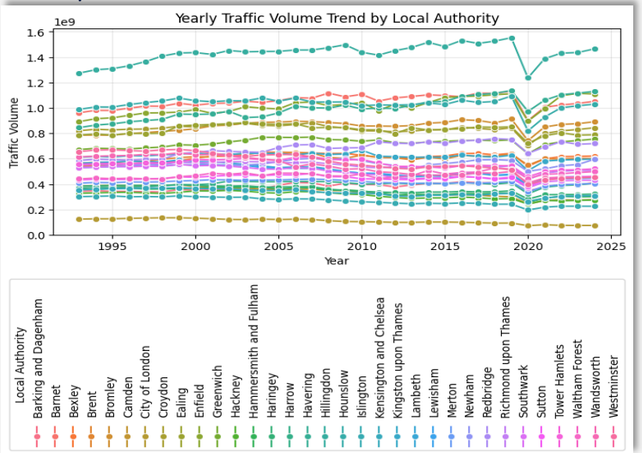

Figure 1: A line graph of yearly traffic volume trend

Why this matters?

London is characterised by unresolved congestion, disproportionate post-COVID recovery, and the growing strain on its road infrastructure. Identification of the location of concentration of traffic and its transformation is important to effectively invest, reduce emissions and in planning operations. It has been documented that data-based corridor targeting gives much greater returns compared to network-wide interventions (Jiang & Luo, 2022).

Figure 6: Tableau Dashboard Presentation

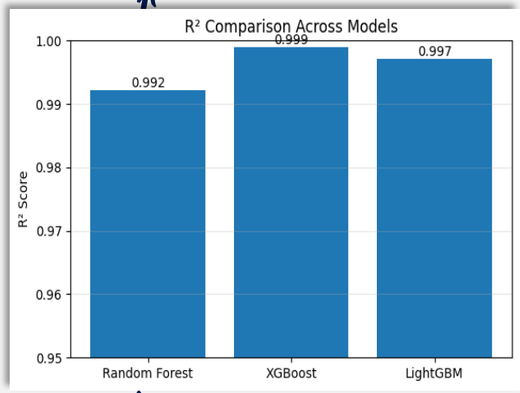

Figure 4: R^2 comparison across models

Data and methods

- Four DfT data sets were incorporated into a single geospatial environment.
- Spatial joins, metre-precise, with EPSG: 27700 Count points equated to MRDB links.
- Hotspots clustering, YoY change detection, and borough-level trend analysis.
- 2019-2024 pattern testing (H1-H3).
- Walk-forward CV for Machine-learning models (Random Forest, XGBoost, LightGBM).
- SHAP explainability to find important predictive drivers.

Who benefits?

- Local Authorities: Justification of the issues of road maintenance, road signal timing, and road congestion strategies.
- Transport for London (TfL): Borough-level recovery insights to assist with network optimisation.
- DfT Policy Teams: Verification of the national trends in traffic and localised behavioural change.
- Emergency Services: Location of high flow corridors of importance to response time.
- Businesses and Logistics Operators: Better knowledge of the demand corridors and bottlenecks.
- Residents and Commuters: They have the potential to decrease delays and enhance reliability of travel.

Findings

The effect on artists, labels, and streaming services: Artist compensation, platform integrity, and revenue distribution.
Ethical issues: Data-driven processes must address privacy concerns, data security, and openness.

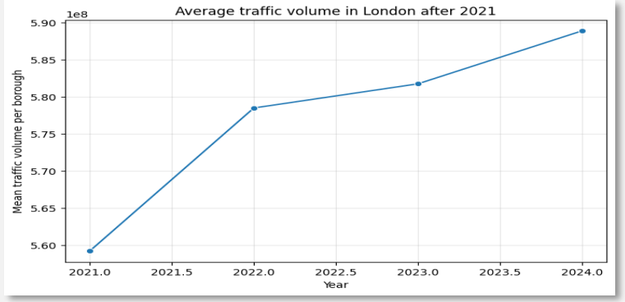

Figure 5: The graph of average volume in London after 2021

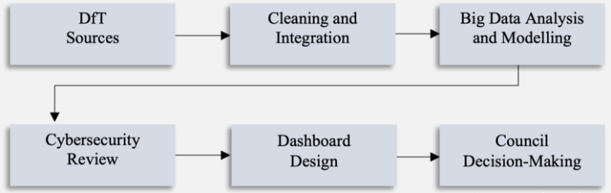

Figure 3: Data Flowchart

Business-related insight

System-wide improvements can be achieved by going after the top 10-15% of high-impact corridors.
Boroughs with rapid-recovery can need a prompt signal retiming and congestion control.
The incorporation of weather, incident and public-transport data would make predictions more accurate.
SHAP-based explainability facilitates evidence-based decision making.

Next steps

- System-wide improvements can be achieved by going after the top 10-15% of high-impact corridors.
- Boroughs with rapid-recovery can need a prompt signal retiming and congestion control.
- The incorporation of weather, incident and public-transport data would make predictions more accurate.
- SHAP-based explainability facilitates evidence-based decision making.

References

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